

**Assessing the Quality and Potential for Sustainable Reuse of Dredged Sediments from
Baltimore Harbor and the Chesapeake Bay**

Natalie Zakamska

On-Site Research Practicum, The Ingenuity Project

Baltimore Polytechnic Institute

1400 W Cold Spring Ln, Baltimore, MD 21209

Contents

Acknowledgments	2
Abstract	2
Introduction	3
Research Focus	5
Sample Sites and Methodology	7
Sample Collection	7
Processing and Analysis	9
Results and Discussion	12
Total Concentrations	12
I_{geo} and mERMQ	13
Correlation Matrix and Leached Mn	15
Screening Values	16
Conclusion	19

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Abstract

Around 4.5 million cubic yards of sediment are dredged from Maryland waterways annually. Dredged material (DM) can be reused for purposes such as land amendment, but contamination prevents using sediments from some areas. Given the quantity of DM and limited storage capacity, we need to understand the risks of reusing DM to make use of this resource. This project analyzed the total concentrations of 4 metals (chromium, copper, manganese, and lead) in sediment samples from sites near Baltimore, and simulated the release of metals from oxidized DM into rainwater. To evaluate contamination and risk, we used the geoaccumulation index (I_{geo}), mean effects range median quotients (mERMQ), and percent differences between experimental concentrations and screening values. We found most Curtis Bay and Inner Harbor sites had higher concentrations of Cr, Cu, and Pb than the main Chesapeake Bay. I_{geo} values showed moderate to strong Cr, Cu, and Pb contamination in most sites, while mERMQ values indicated $\geq 50\%$ risk for most sites and $> 80\%$ risk for the most contaminated site. Mn concentrations exceeded several guidelines despite being within natural ranges. A high Mn concentration in the deeper Chesapeake sample indicates that geochemical cycling may influence the contamination of DM. However, concentrations of Cr, Cu, and Pb were typically below EPA industrial RBCs and the ITRT non-residential reuse crite-

ria by 50 – 100%, showing a possibility for industrial reuses. Future research should analyze the chemical changes of DM over time, and the influence of other variables and exposure pathways on actual toxicity to organisms.

Introduction

Heavy metals (HMs) are a major source of water and soil pollution globally. Some HMs are essential to human and animal life at the micronutrient level, while others are toxic even in trace amounts (Dagdag et al., 2023; Moes et al., 2019b). Chronic exposure to HMs may cause tumors, osteoporosis, issues with the cardiovascular and gastrointestinal systems, and developmental impairments, especially in young children (Dagdag et al., 2023). In aquatic organisms, metal toxicity has numerous effects: it decreases the abundance and diversity of species; changes reproduction, juvenile growth, and behavior; and causes abnormalities and death. The processes of bioaccumulation and biomagnification, in which metals accumulate in organisms and their predators more quickly than they can be expelled, can cause metal concentrations to be toxic in organisms and to the humans or other wildlife that consume them (Environmental Protection Agency [EPA], 2021).

Because their properties make them useful in industry, pollution has caused HMs to become much more prevalent in water, soil, and to a lesser extent, air. Moreover, because metals are not biodegradable, they may remain in the environment for extended periods of time until conditions allow them to mobilize and be taken up by organisms, although oxidation state and speciation (the form of the metal in the environment) may decrease HM toxicity (Dagdag et al., 2023; Moes et al., 2019a). Sediment especially is a contamination concern because metals accumulate in aquatic environments, causing them to have the largest concentrated pool of HMs, which may be released upon disturbance (Acevedo-Figueroa et al., 2006; Dagdag et al., 2023). Knowing this, it is important to assess pathways of exposure to HMs, especially in areas with currently or previously active industry, where the highest concentrations of HMs are present (Dagdag et al., 2023).

An example of such a location is Baltimore, Maryland, a port city that has had major industrial activity throughout its history, including ore mining and processing, canning, textile manufactur-

ing, ship and plane building, and other metal-intensive industries like paint making and ammunition production (Mason et al., 2004). Investigations into the metal concentrations of sediments in waterways surrounding Baltimore have established that HM input and levels have decreased compared to historical times, but continue to be elevated, especially in certain areas (Baker et al., 1997; Mason et al., 2004). This has the potential to impact populations of humans and other organisms because these waterways must be dredged regularly to maintain depth for ships to pass safely, which involves numerous routes of exposure depending on how the DM is processed and used (Kramer & Greer, 2009; The Independent Technical Review Team [ITRT], 2009). Baltimore Harbor alone requires 1.5 million cubic yards of sediment to be removed annually—enough to pave an 8 foot by 4 inch driveway from Baltimore to San Francisco every year (Kramer & Greer, 2009).

DM can be reused in certain ways. Maryland defines DM reuse as beneficial when used in habitat restoration, and innovative when it is used in developing and manufacturing industrial, commercial, and agricultural products (ITRT, 2009). However, the volume and contamination of this dredged material (DM) due to human activities that release HMs into sediment make it challenging to dispose of or manage safely and effectively. According to the Dredged Material Management Plan (DMMP) created by US Army Corps of Engineers Baltimore District (USACE, 2017), harbor sediments (defined as inside the North Point-Rock Point Line) are legally contaminated and generally cannot be used beneficially (i.e. in habitat restoration); they must be contained. Unfortunately, Baltimore Harbor faces a shortage of DM storage due to the limited number and capacities of Dredged Material Containment Facilities (DMCFs). Several have been closed due to reaching capacity since 2005, when the DMMP was created (Kramer & Greer, 2009).

With a finite storage capacity for DM in the harbor and safety standards to be met for types of reuse, there have been investigations on how best to use DM (Kramer & Greer, 2009; ITRT, 2009; USACE, 2017). In 2008, the Maryland Port Administration formed an Independent Technical Review Team (ITRT) to determine the suitability of sediment from the harbor and the surrounding area for innovative reuse. The team conducted a comprehensive analysis of 20 metals and over 100

organic contaminants in numerous samples spanning a range of years. They examined pathways of risk and compiled existing guidelines to develop screening protocols for four types of land reuse (listed in descending order of criteria strictness): unrestricted, residential, non-residential, and unsuitable for use. They found that most locations met criteria necessary for non-residential (i.e. industrial) use, while fewer met residential criteria and none met unrestricted criteria. However, considering that the team's decision matrix cannot provide specific enough guidance for individual projects, the team recommended a case-by-case assessment of DM using chemical analysis and the compiled criteria for the specific analytes and appropriate reuse option. Additionally, not all possible reuse options/risk pathways were evaluated (such as ones that involved more complex exposures), certain contaminants were missing from the data, and comparison between the Inner Harbor and Chesapeake Bay was limited (ITRT, 2009).

Research Focus

Sediment quality and HM contamination throughout Baltimore Harbor, the Patapsco, and the Chesapeake have been thoroughly assessed, although with some limitations. Inadequate management of both sewage and industrial waste, as well as leaching from industry, have contributed to the continued elevation of metal levels in sediments of the harbor today (Mason et al., 2004). While metal concentrations are much improved from the previous two centuries, most locations in the harbor and along the Patapsco exhibit elevated levels of metals such as Cr, Cu, Pb, and Zn. The Inner Harbor, Curtis Bay, and Bear Creek are just a few locations where a majority of sediment samples taken have metal concentrations that would cause low to moderate toxicity for organisms (Baker et al., 1997). Existing guidelines have been consolidated and evaluated for use in DM operations, and screening processes have been created to determine the use of DM. However, this still requires case-by-case assessments (Kramer & Greer, 2009; ITRT, 2009).

This project determines the total concentrations of several HMs in sediment samples from the Inner Harbor and Chesapeake Bay, and compares concentrations to established guidelines to assess potential reuse options, risk, and need for further investigation. Specifically, it seeks to identify the

most fitting categories of reuse for these samples. The analysis comprises four metals: chromium (Cr), copper (Cu), manganese (Mn), and lead (Pb). All four metals are associated with the region; Cr, Cu, and Pb are enriched in harbor sediments due to historical and current sources (Mason et al., 2004), while Mn is known to occur at high concentrations naturally in Maryland (ITRT, 2009).

Additionally, this project examines the spatial variation of sediment contamination and, therefore, reuse potential. HM concentrations are known to be higher in the Inner Harbor than the Chesapeake Bay, therefore constraining reuse options for harbor sediments, although high concentrations do not necessarily prohibit reuse (ITRT, 2009). The influence of realistic environmental conditions (such as oxidation and rain) and qualities of the sediment (such as carbon) on the release of metals was also examined using an elutriate test, where dried sediment was mixed with rainwater and allowed to settle before the supernatant was analyzed for leached metal concentrations.

This project also presented several predictions:

- There would be a positive correlation between Cr, Cu, and Pb levels due to a common source (anthropogenic activity) but little correlation with Mn due to its high natural occurrence.
- The order of contamination, from lowest to highest, would be the Chesapeake Bay, then the Inner Harbor, and lastly, Curtis Bay as a result of industrial activity.
- Some portion of all four metals would leach into rainwater, although not the whole amounts, reflecting the presence of metals in the more tightly bound oxidizable and residual fractions (Feng et al., 2023).
- There would be a positive correlation between Cr, Cu, and Pb concentrations and carbon (C) as metals tend to be concentrated in the organic C fraction (Mason et al., 2004).

Sample Sites and Methodology

Sample Collection

The sediment samples used in this project were collected from different locations with varying methods. Sediments from the Inner Harbor near the National Aquarium and Fort McHenry (NA and FM sites) were collected in August 2025. Sediments from Curtis Bay (CUR sites) and the Chesapeake (CB sites) were collected by Dr. Maya Gomes at Johns Hopkins University in March 2025 and July 2019 respectively. Within the CB site, 2 samples were taken: one from surface sediment, 3.5 cm below the surface of the sediment (CB 3.3W - 3.5), and one from depth sediment, 47.5 cm below the sediment surface (CB 3.3W - 47.5). All other samples were of surface sediment. The locations of the sample sites and their coordinates are given below in Figure 1 and Table 1. Curtis Bay is technically within the Inner Harbor borders, and was considered as such in analysis. However, CUR sites were also considered separately from the NA and FM sites as Curtis Bay is subject to several unique sources of HM pollution, including the Coast Guard Yard and a history of heavy manufacturing.

Sample ID	Coordinates
CUR 1	39.21876N, 76.57736W
CUR 2	39.22475N, 76.57803W
CUR 3	39.22519N, 76.57053W
CUR 4	39.22278N, 76.56905W
CUR 5	39.21809N, 76.56322W
CUR 6	39.21972N, 76.57035W
FM K0825	39.25802N, 76.58468W
NA K0824	39.28508N, 76.61058W
CB 3.3W (3.5 and 47.5)	39.00468N, 76.38967W

Table 1: Sample IDs and coordinates.

The NA and FM samples were obtained using a stainless steel Van Veen grab sampler. The sediments were emptied into a plastic bucket rinsed with bottom waters from the same location. Large visible organic material was removed from the sediment so it would not impede sample collection. Sediment in contact with the Van Veen sampler was removed with acid-washed spoons.

Sediment was then scooped into 50mL centrifuge tubes and quickly capped with no headspace. After cleaning with paper towels, the tubes' caps were sealed with parafilm and the tubes were stored in the dark.

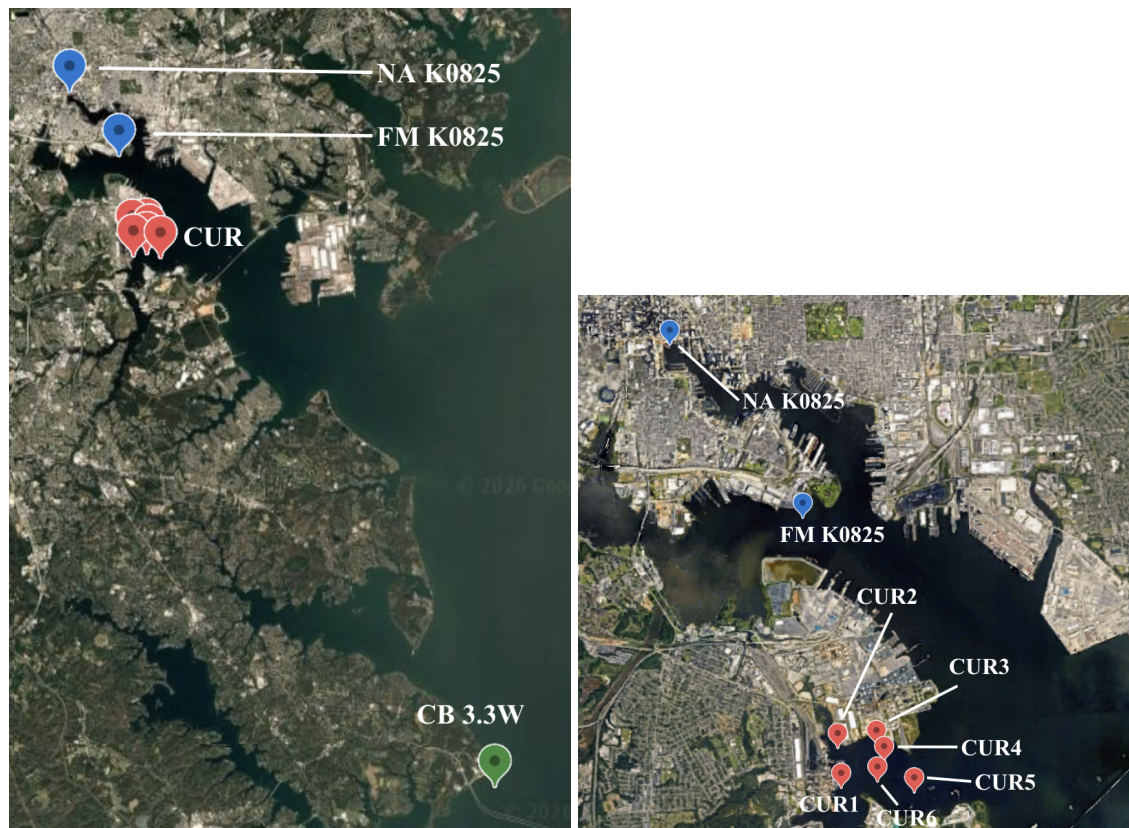


Figure 1: Left: Locations of all sample sites. Right: NA, FM, and CUR sites only.

Sampling for elemental concentrations was done using nitrile gloves and materials cleaned for trace element grade analysis according to the acid washing procedures of Johannesson et al. (2004). This process includes multiple repetitions of triple washing tubes, plastic spoons, and bags with deionized water and placing them in trace metal grade acid baths before air drying and storage.

Upon returning to the laboratory at Johns Hopkins University the same day, the samples were centrifuged (25 min at 4200 RPM) to separate porewater from the sediments and stored with nitric acid. Sample handling was performed in a nitrogen-purged glove bag. Sediment samples were sealed in foil barrier bags and stored at -20°C .

A detailed description of the process (including trace metal clean procedures) used to collect

samples from Curtis Bay can be found in Cui et al. (2021). Samples were obtained using a multicorer and processed in a glove bag with a nitrogen atmosphere immediately after retrieval. For the Chesapeake Bay site, the methodology (non trace metal clean procedures) can be found in Hantsoo et al. (2024). Chesapeake samples were collected with a gravity corer and also processed in a nitrogen-purged glove bag where they were separated according to sediment depth and sealed in foil barrier bags at -20°C .

Rainwater was collected in a plastic bin conditioned with sample water, then placed into plastic bottles also conditioned with the sample water. The presence or lack of trace metal clean procedures was not the primary limitation of this project and was therefore not considered a major factor in data analysis.

Processing and Analysis

Upon retrieval of the samples from the laboratory at Johns Hopkins, all samples were processed in the same manner. Tubes were removed from foil barrier bags and uncapped in a nitrogen atmosphere glove bag. Approximately 3cm^3 of sediment from each location was loaded into syringes using metal spatulas before being placed in aluminum weigh boats. To ensure no cross contamination between samples, all materials were cleaned with deionized water and Kimwipes. After processing, the tubes were re-capped, parafilm, and placed back into foil barrier bags. The weigh boats were then placed in an oven to dry at 60°C for several days.

After drying, sediments were powdered using a marble mortar and pestle and double bagged in brand new plastic bags. Care was taken to ensure minimal cross contamination: materials were cleaned with water and 70% isopropyl alcohol between samples, and fresh paper towels were placed on the work area.

After being shipped to University of San Diego, the samples were analyzed for total metal concentrations, CHNS as a proxy for organic matter, and metals leached into the supernatant in the rainwater test. For samples with sufficient amounts to complete the analysis multiple times, duplicates were run. To ensure quality control throughout the process, blanks were run for rainwater

and total digestion analysis, and NIST standard reference materials were run for total digestion and CHNS analysis. All values reported were confidently above the detection limits of the instruments.

Total digestion was completed according to EPA Method 3051 (microwave assisted nitric acid digestion). Approximately 0.5g of each sample were added into microwave vessels along with 10mL of concentrated nitric acid. Weights were recorded to 3 decimal places. After the microwave program was run and samples were allowed to cool, they were placed in 50mL centrifuge tubes and analyzed by ICP-OES.

For the elutriate test, approximately 0.5g of each sample were placed into 50mL centrifuge tubes along with 25mL of rainwater which were pipetted into the tubes. The tubes were inverted several times and then allowed to settle. Concentrated nitric acid was added before ICP analysis.

To analyze for CHNS, 4-50mg of each sample was weighed out before being placed in the analyzer. Some samples did not have sufficient material to run duplicates, and in some cases there was not enough to test for CHNS at all.

After data collection, data analysis was completed in Excel. Excel was used to create a Pearson correlation coefficient matrix, as significant correlation between elements may indicate a common origin (Acevedo-Figueroa et al., 2006). For samples with duplicates, weighted averages for elemental concentrations were calculated using the masses of the samples. Percent differences between experimental values and screening values were calculated and graphed in Excel, treating screening values as a baseline according to the following equation:

$$\% \text{difference} = \frac{C_m - S_m}{S_m} * 100\%, \quad (1)$$

where C_m is the concentration of the metal in the sample and S_m is the screening value for the metal.

Most screening/cleanup values and quality standards/criteria for various media (such as sediment and soil) were used as compiled in ITRT (2009). These included EPA Region 3 Risk-Based Concentrations (RBCs) for soil, the ITRT non-residential reuse criteria for soil, EPA Ecological Soil Screening Levels for plants, and the Apparent Effects Threshold (AET). Values not mentioned

in ITRT (2009) included the Biological Technical Assistance Group (BTAG) freshwater screening benchmarks and the anticipated typical concentration (ATC) for the studied HMs in soil in central Maryland.

Other calculated measures of risk and contamination included the geoaccumulation index (I_{geo}) and mean Effects Range Median quotients (mERMQ). I_{geo} is calculated as follows:

$$I_{geo} = \log_2\left(\frac{C_n}{1.5B_n}\right), \quad (2)$$

where C_n is the concentration of the metal in sample sediment and B_n is the concentration of the metal in the Earth's crust.

The Effects Range Median (ERM) sediment quality guideline is one of several compiled by the National Oceanic and Atmospheric Administration (NOAA) into Screening Quick Reference Tables (SQuiRTs) as preliminary screening purposes for marine, estuarine, and freshwater sediments (Buchman, 2008; Ramos et al., 2021). The ERM guideline for a chemical is defined as the 50th percentile of concentrations from effects data sets of toxicity tests and benthic community surveys and is intended for marine sediment (Buchman, 2008; ITRT, 2009). The ERMQ method assesses the degree of HM contamination according to the toxicity in marine organisms by expressing the sample concentration as a fraction of the ERM value. The mERMQ value considers the ERMQ values of multiple metals and is calculated according to the following equation:

$$ERMQ_i = \frac{c_i}{ERM_i}, \quad (3)$$

$$mERMQ = \left(\sum_{i=1}^n ERMQ_i\right)/n, \quad (4)$$

where c_i is the concentration of the studied metal in sample sediment, ERM_i is the ERM reference value according to the NOAA SQiRTs (Buchman, 2008), and n is the total number of metals analyzed at the site. Ramos et al. (2021) considered the risk percentages according to mERMQ values to be relative to the probability of toxicity in amphipod survival bioassays. Mn was not

considered in mERMQ calculations, as there is no ERM value provided in Buchman (2008).

Results and Discussion

Total Concentrations

Sediments from the Inner Harbor and Chesapeake Bay show spatial differences. Concentrations of Cr, Cu, and Pb were higher in nearly all of the Inner Harbor sites than either the surface or depth samples from CB 3.3W. However, contamination in the CB sample increased with depth. Of the Inner Harbor sites, the NA site had the highest concentrations of Cu, Cr, and Pb, while CUR2 had the lowest concentrations. The Cr concentration of CUR2 was lower than either the surface or depth CB sample, and the CUR2 Pb concentration was also lower than CB 3.3W - 47.5. The remaining CUR sites and the FM site had concentrations between those of the CB sites and the NA site.

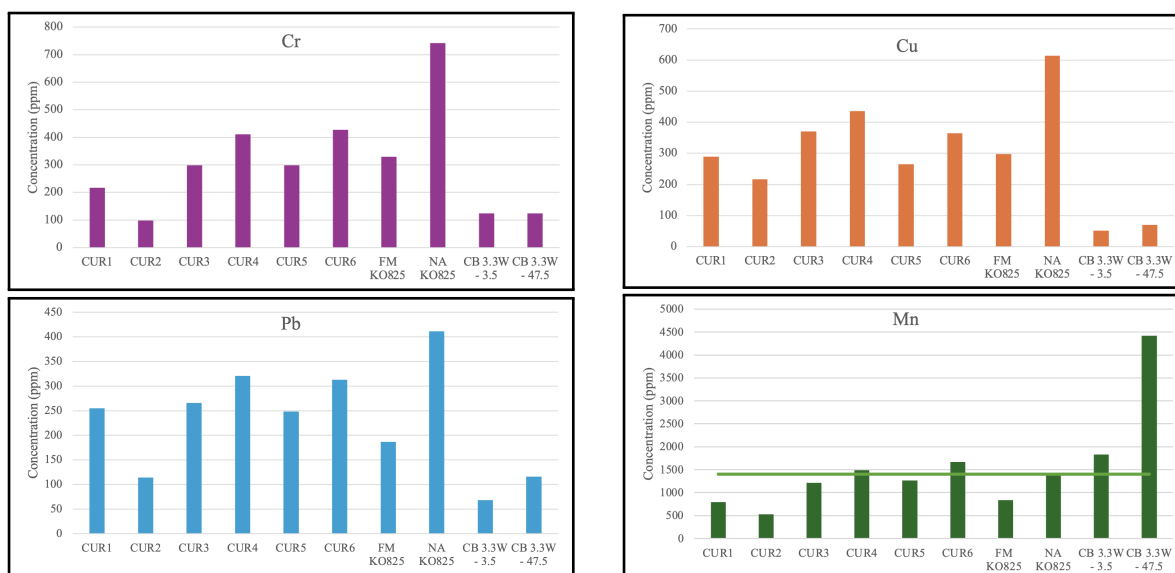


Figure 2: Total concentrations (ppm) of studied metals across the sample sites. The horizontal line in the graph for Mn represents the anticipated typical concentration (ATC) for Mn in central Maryland.

Regarding Mn, 6 sites were below the ATC for central Maryland (1400ppm). 3 sites exceeded it by less than 500ppm. However, the depth sample for CB 3.3W had a concentration of Mn far

higher than any other sample (over 4400ppm). The disparity between Cr, Cu, and Pb (metals most likely impacted due to anthropogenic sources) can likely be explained due to proximity to shore (and therefore sources of HM), and presence or lack of dredging, which can remove overlying contamination. NA is likely too close to shore to be subject to dredging, whereas CB is located farther from major HM sources and may be influenced by dredging operations.

The unusually high value of Mn in CB 47.5 may be a result of legacy inputs or, more likely, the geochemical cycling of Mn. The mid Chesapeake Bay has seasonal anoxia and high sulfide content in bottom sediments, which reduces Mn oxides, causing them to become the soluble form Mn^{2+} . In the presence of sulfur or iron sulfides, Mn is often incorporated into sulfur-containing compounds, which may explain the high presence of Mn in deeper sediments. However, more in-depth study of the environmental conditions of this sample would be necessary to draw conclusions.

I_{geo} and mERMQ

The geoaccumulation index was calculated to assess the degree of contamination for the studied metals, while mERMQ was calculated to estimate the percentage of risk to sediment infauna as a proxy for the toxicity of the sediments. These levels of contamination and risk are shown below in Tables 2 and 3, while the I_{geo} and mERMQ values for different sites are shown in Figures 3 and 4.

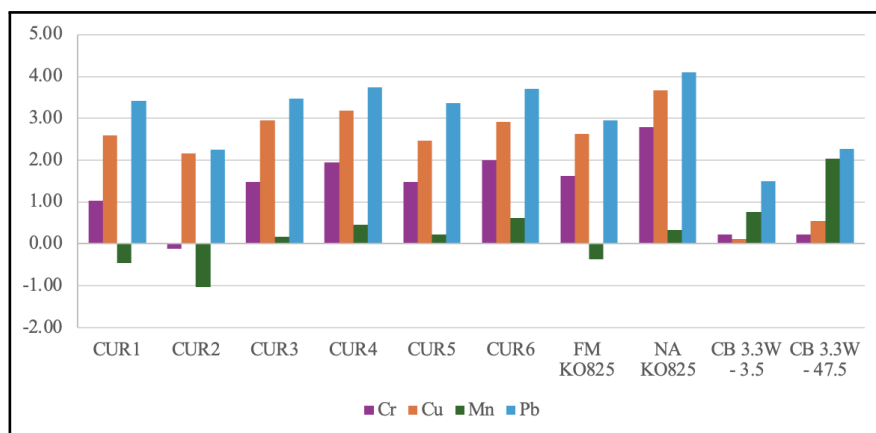


Figure 3: Geoaccumulation index (I_{geo}) values across the sample sites.

Using the scale in Table 2, the I_{geo} values for all sites indicate moderate to strong Pb and Cu contamination. For Cr, almost all sites exhibit moderate contamination with the exception of

Value	Classification
$I_{geo} \leq 0$	Uncontaminated
$0 < I_{geo} < 1$	Uncontaminated to moderately contaminated
$1 < I_{geo} < 2$	Moderately contaminated
$2 < I_{geo} < 3$	Moderately to strongly contaminated
$3 < I_{geo} < 4$	Strongly contaminated
$4 < I_{geo} < 5$	Strongly to extremely contaminated
$5 < I_{geo}$	Extremely contaminated

Table 2: Contamination categories based on I_{geo} .

ERMQ	Percentage of risk
< 0.1	12
$0.11 - 0.5$	30
$0.5 - 1.5$	46
> 1.5	76

Table 3: Risk percentages based on ERMQ values.

CUR2 which is uncontaminated. Mn also shows low to no contamination, except in CB 3.3W - 47.5, where it is moderately contaminated.

As is consistent with the total concentrations, the NA site had the highest I_{geo} values for Cr, Cu, and Pb. Across all sites, Pb had the highest I_{geo} value, which indicates that it is the most contaminated relative to natural concentrations and that this contamination is relatively uniform throughout the harbor.

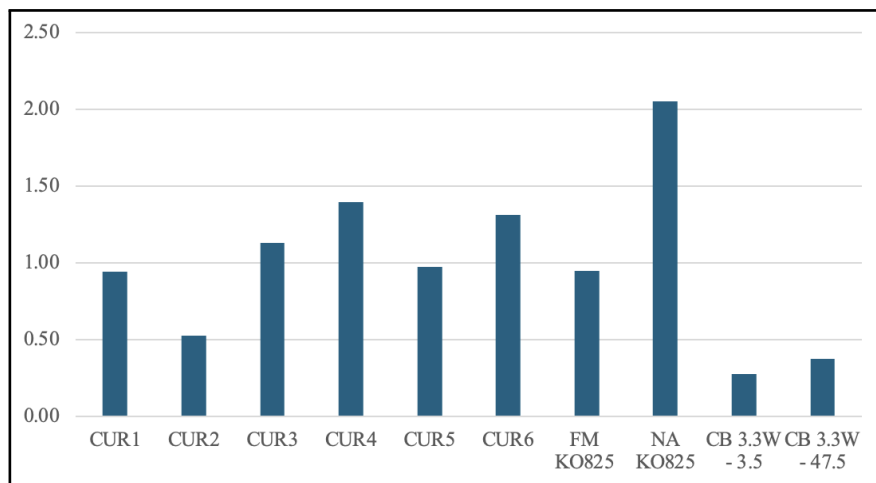


Figure 4: mERMQ values across the sample sites.

Seven out of ten sites had mERMQ values between 0.5 and 1.5, showing a $\geq 50\%$ risk for most sites and a $> 80\%$ risk at NA site based on Cr, Cu, and Pb. Especially for NA, this may indicate toxicity to organisms and therefore that sediments should be reused with caution.

Correlation Matrix and Leached Mn

A correlation matrix was created in Excel using the Pearson correlation coefficient (r), which indicates the strength and direction of a linear relationship between two variables. It showed several surprising relationships which contradicted the predictions of this project, especially regarding organic matter (which was analyzed using CHNS as a proxy) and what the metals were associated with.

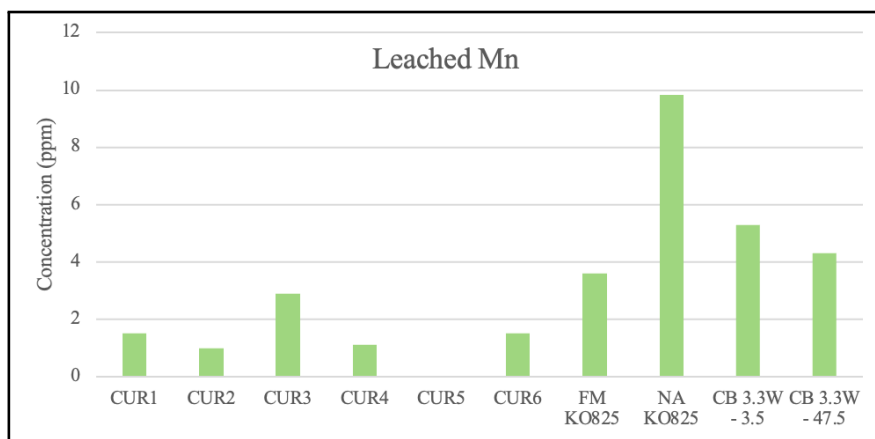


Figure 5: Concentration of Mn (ppm) in the supernatant of the elutriate test.

In the elutriate test, very little Mn leached (much less than the 120ppm BTAG surface water guideline) and no detectable amounts of other metals leached, indicating that metals are tightly bound to the sediment matrix. Furthermore, the lack of a strong negative correlation between leached Mn and C, N, or S (which may have indicated that organic matter was preventing Mn from becoming mobile) likely indicates that other mechanisms control mobility of Mn in these sediments.

However, strong correlations between Cr, Cu, and Pb indicate a likely common source (anthropogenic input) (Acevedo-Figueroa et al., 2006), while weak correlations with Mn show that Mn is

	<i>Cr</i>	<i>Cu</i>	<i>Mn</i>	<i>Pb</i>	<i>Leached Mn</i>	<i>C [%]</i>	<i>N [%]</i>	<i>S [%]</i>
Cr	1							
Cu	0.92	1						
Mn	-0.20	-0.43	1					
Pb	0.91	0.94	-0.27	1				
Leached Mn	0.50	0.28	0.24	0.19	1			
C [%]	0.06	0.40	-0.50	0.27	-0.23	1		
N [%]	0.57	0.72	-0.31	0.73	0.18	0.69	1	
S [%]	0.73	0.78	-0.32	0.87	0.27	0.30	0.64	1

Figure 6: Correlation matrix of metal concentrations and elemental percentages. Values closer to +1 (dark blue) or -1 (dark red) indicate stronger linear relationships between variables.

likely geogenic, which is consistent with the initial predictions of this project. Although correlations between Cr, Cu, and Pb and C were typically weak (< 0.5), stronger correlations with N and S show that those two elements may constitute a greater portion of the organic matter associating with the metals and preventing them from leaching.

Screening Values

While several screening values, standards, and criteria were used as comparisons to the concentrations of HMs in this project, it is important to note that many are intended for preliminary screening purposes and do not constitute an official level for cleanup. Instead, they are meant to assess if further remediation is necessary. Percent differences were calculated using the AET (Figure 7), the ITRT non-residential reuse criteria (Figure 8), EPA Region 3 Soil RBCs (Figure 9), and EPA Ecological Soil Screening Level (EcoSSL) for plants (Figure 10).

As seen in Figure 7, Cu and Pb levels for the sites are generally lower than the AET value, although not by much. Because the AET value represents the threshold above which adverse effects are expected (ITRT, 2009), values below the AET still represent potentially toxic levels of exposure. Cr and Mn levels higher than AET values indicate risk to organisms, as this means adverse effects are expected.

Cu and Pb concentrations for all sites were also lower than the ITRT non-residential reuse criteria by 50–100%, showing a promising case for industrial reuses. The ITRT non-residential reuse criteria is derived from and is, in many cases, the same as the Maryland soil and ground-

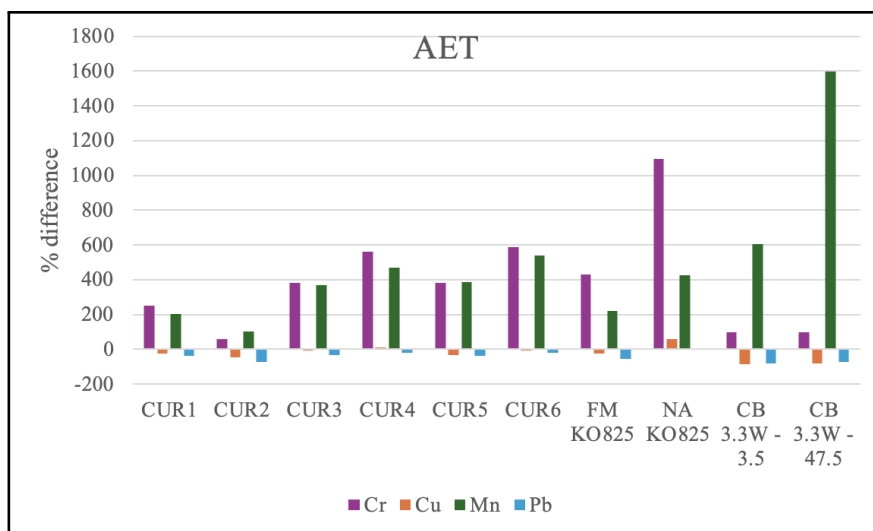


Figure 7: Percent differences between concentrations of metals at sample sites and the AET sediment quality guideline.

water cleanup standards. Therefore, it provides a good reference point for the official levels that contaminants must be below to qualify for reuse. However, Cr and Mn for several sites exceeded this criteria, showing a constraint on industrial reuses.

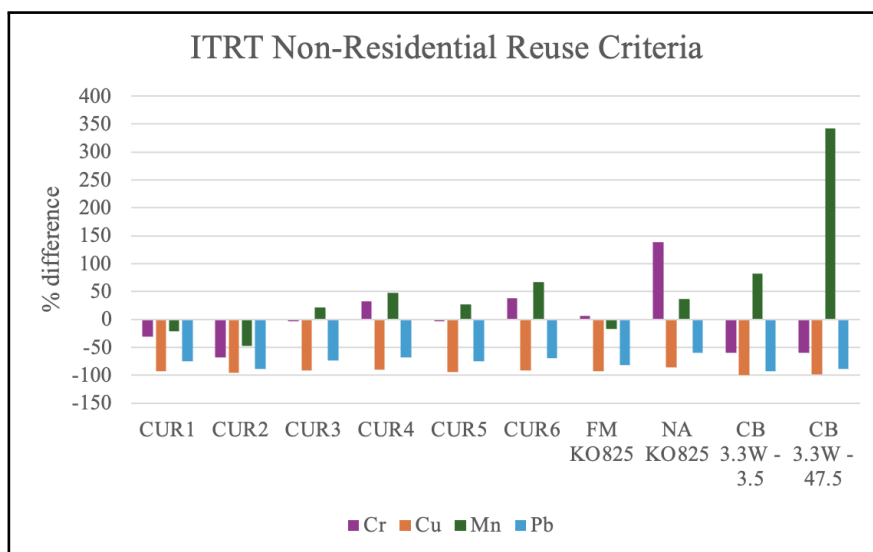


Figure 8: Percent differences between concentrations of metals at sample sites and the ITRT non-residential reuse criteria.

Similarly, as shown in Figure 9, Cu and Pb were well below the RBCs. However, Mn levels exceeded the EPA RBCs less than they exceeded the ITRT's criteria.

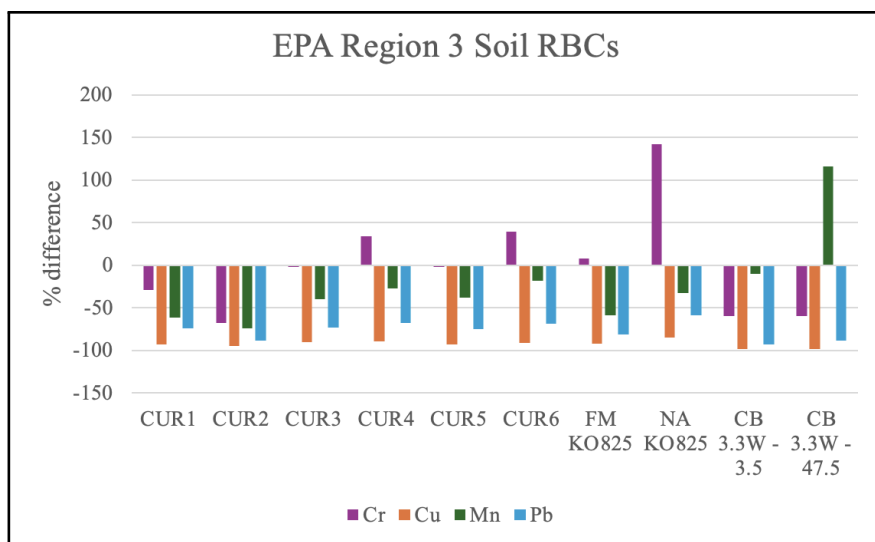


Figure 9: Percent differences between concentrations of metals at sample sites and the EPA Region 3 RBCs for soil.

Because the concentrations of the studied metals also exceed the EcoSSL value in nearly all sample sites by hundreds of percentage points, as shown in Figure 10, there is a case for limiting the application of DM from these locations to uses that will not involve plant growth (e.g. residential reuses). However, the fact that Mn is shown to exceed this screening value despite being within natural ranges reflects a need for more local screening values which consider natural concentrations in their derivation.

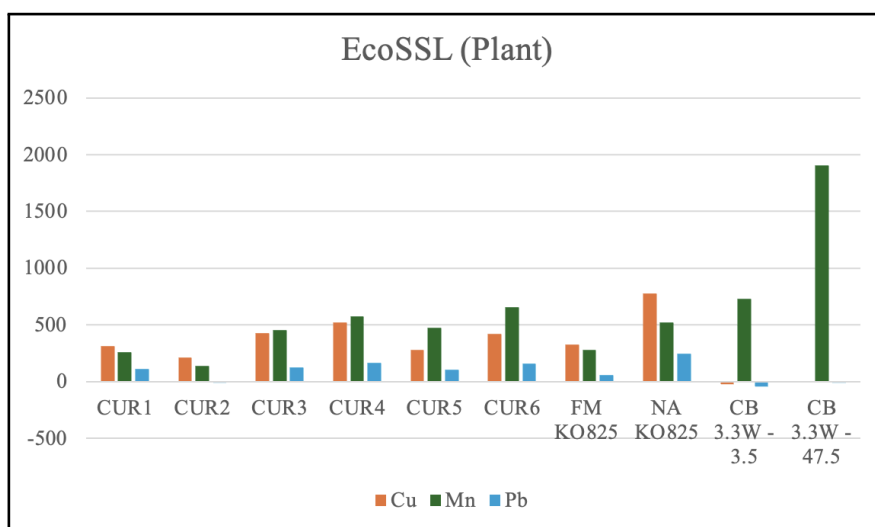


Figure 10: Percent differences between concentrations of metals at sample sites and the EPA EcoSSLs for plants.

Overall, Cu and Pb levels are generally lower than industrial screening values (Figures 8 and 9). This shows a possibility for industrial reuse. Although Cr and Mn are the most limiting factors according to the ITRT non-residential reuse criteria, there are several sites for which the percent difference is less than 100. According to ITRT (2009), sediments with concentrations exceeding non-residential guidelines may still be reused in innovative reuses, granted that the end use and engineering and institutional controls are appropriate. However, the risk of toxicity indicated by the exceedance of the AET and EcoSSL values shows a limitation in reuse which needs to be investigated further. The selected screening values are primarily applicable to DM reused as soil, so use of further testing, other criteria, and simulations of other risk pathways are necessary to draw conclusions about other industrial reuses.

Conclusion

Some harbor sediments are very contaminated and should not be reused due to high risk and potential toxicity. Others might be considered for industrial reuses, although this is limited by continued contamination from sources like industry and wastewater. Main Chesapeake sediments are less contaminated and therefore have a wider variety of possible reuse options, but they should still be assessed for risk, especially deeper sediments which may be affected by historical contamination (similar to Inner Harbor sediments) or the geochemical cycling of HMs.

Although this project was limited by its small scale, both in terms of number of samples and variables analyzed, the results show that the studied metals are tightly bound to the sediment matrix and are likely not an immediate threat. However, the reliance on previous guidelines prevents concrete conclusions from being drawn about the actual toxicity of these samples if they were reused as DM. Further research is necessary to identify the specific factors affecting metal mobility and bioavailability in these sediments, including speciation, acid volatile sulfide (AVS), and a wider variety of chemical components. The current possibilities for reuse options using harbor and bay sediment would also benefit from chemical analysis and direct toxicity tests of samples in natural conditions over time to better understand long-term implications of DM use and beneficial or detri-

mental chemical changes. Additionally, a comprehensive review of the accuracy and limitations of current criteria, especially for metals such as Mn which may naturally occur in high concentrations, is necessary. Finally, knowing the compatibility for a greater number of exposure pathways, which may require complex testing, is another important step to assessing the reuse potential of DM from the harbor and bay.

Ultimately, knowing what local DM can be used for and what limitations exist for DM from certain locations will help us more effectively make use of it. In this way, we can help create a more sustainable society by drawing from a unique and thus far, relatively untapped, material.

References

- Acevedo-Figueroa, D., Jiménez, B., & Rodríguez-Sierra, C. (2006). Trace metals in sediments of two estuarine lagoons from Puerto Rico. *Environmental Pollution*. <https://doi.org/doi:10.1016/j.envpol.2005.08.037>
- Baker, J., Mason, R., Cornwell, J., Ashley, J., Halka, J., & Hill, J. (1997). Spatial mapping of sedimentary contaminants in the Baltimore Harbor/Patapsco River/Back River system. *Maryland Department of the Environment*. <https://aquadocs.org/items/4ca50a12-b4a8-4ed4-b2a3-aa392a6d68e0>
- Buchman, M. F. (2008). *Screening quick reference tables (squirts)* (tech. rep.). NOAA.
- Cui, M., Luther III, G. W., & Gomes, M. (2021). Cycling of W and Mo species in natural sulfidic waters and their sorption mechanisms on MnO₂ and implications for paired W and Mo records as a redox proxy. *Geochimica et Cosmochimica Acta*, 295, 24–48. <https://doi.org/https://doi.org/10.1016/j.gca.2020.12.007>
- Dagdag, O., Quadri, T. W., Haldhar, R., Kim, S.-C., Daoudi, W., Berdimurodov, E., Akpan, E. D., & Ebenso, E. E. (2023). An overview of heavy metal pollution and control. In *Heavy metals in the environment: Management strategies for global pollution* (pp. 3–24). American Chemical Society. <https://pubs.acs.org/doi/10.1021/bk-2023-1456.ch001>

- Environmental Protection Agency. (2021). Factsheet on water quality parameters: Metals [Accessed: 2018-12-06].
- Feng, J., Burke, I. T., Chen, X., & Stewart, D. I. (2023). Assessing metal contamination and speciation in sewage sludge: Implications for soil application and environmental risk. *Reviews in Environmental Science and Bio/Technology*, 22, 1037–1058. <https://link.springer.com/article/10.1007/s11157-023-09675-y>
- Hantsoo, K., Gomes, M., Brenner, D., Cornwell, J., Palinkas, C. M., & Malkin, S. (2024). Trends in estuarine pyrite formation point to an alternative model for Paleozoic pyrite burial. *Geochimica et Cosmochimica Acta*, 374, 51–71. <https://doi.org/10.1016/j.gca.2024.04.018>
- Johannesson, K. H., Tang, J., Daniels, J. M., Bounds, W. J., & Burdige, D. J. (2004). Rare earth element concentrations and speciation in organic-rich blackwaters of the Great Dismal Swamp, Virginia, USA. *Chemical Geology*, 209(3), 271–294. <https://doi.org/10.1016/j.chemgeo.2004.06.012>
- Kramer, J. G., & Greer, J. (2009). Sediment in Baltimore Harbor: Summary of an independent technical review.
- Mason, R. P., Kim, E.-H., & Cornwell, J. (2004). Metal accumulation in Baltimore Harbor: Current and past inputs. *Applied Geochemistry*, 19(11), 1801–1825. <https://doi.org/10.1016/j.apgeochem.2004.03.005>
- Moes, S., van Gestel, K., & van Beek, G. (2019a). Metal speciation. In *Environmental toxicology*. LibreTexts. [https://chem.libretexts.org/Bookshelves/Environmental_Chemistry/Environmental_Toxicology_\(van_Gestel_et_al.\)%29%3A_Environmental_Chemistry_-_From_Fate_to_Exposure%3A_Metal_Speciation](https://chem.libretexts.org/Bookshelves/Environmental_Chemistry/Environmental_Toxicology_(van_Gestel_et_al.)%29%3A_Environmental_Chemistry_-_From_Fate_to_Exposure%3A_Metal_Speciation)
- Moes, S., van Gestel, K., & van Beek, G. (2019b). Pollutants with specific properties. In *Environmental toxicology*. LibreTexts. [https://chem.libretexts.org/Bookshelves/Environmental_Chemistry/Environmental_Toxicology_\(van_Gestel_et_al.\)%29%3A_Environmental_Chemistry_Chemicals%2A_Pollutants_with_specific_properties](https://chem.libretexts.org/Bookshelves/Environmental_Chemistry/Environmental_Toxicology_(van_Gestel_et_al.)%29%3A_Environmental_Chemistry_Chemicals%2A_Pollutants_with_specific_properties)

- Ramos, R., Verde, A., & García, E. M. (2021). Heavy metals in Venezuelan marine sediments: Concentrations, degree of contamination, and distribution. *Ciencias Marinas*, 47, 185–199. <https://doi.org/10.7773/cm.v47i3.3124>
- The Independent Technical Review Team. (2009). *Sediment in Baltimore Harbor: Quality and suitability for innovative reuse* (tech. rep.). Maryland Sea Grant.
- US Army Corps of Engineers Baltimore District. (2017). Baltimore Harbor and channels dredged material management plan update.